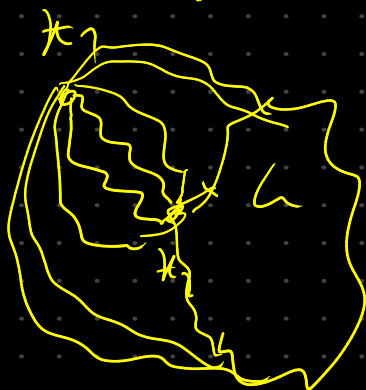


Continuing on proof of Theorem 2.1.

Case where $L - x_1$ sep. into multiple comps.

What we had

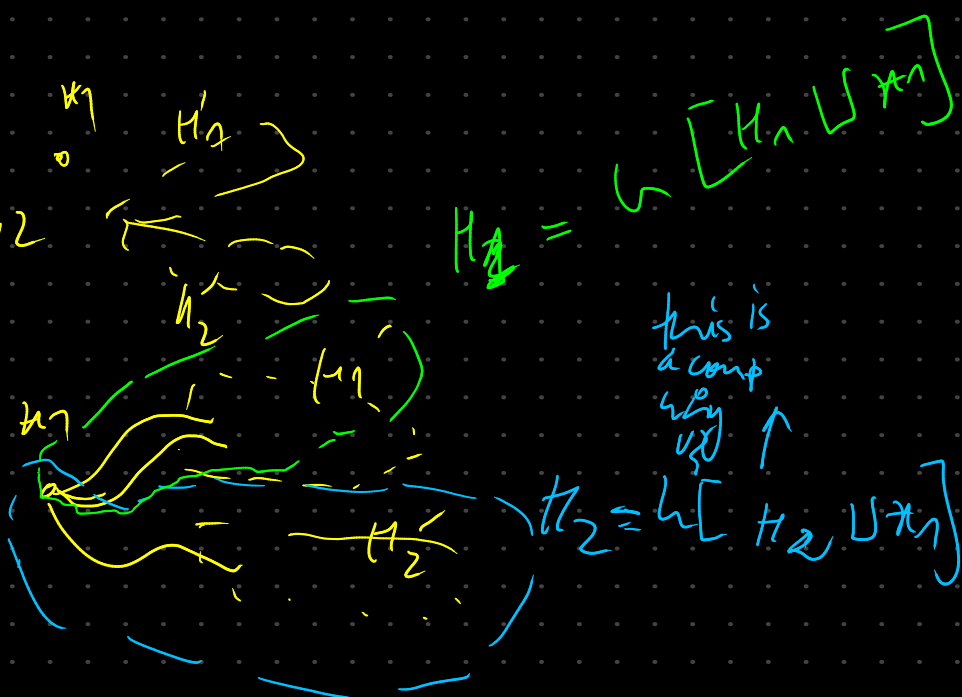


What's left:



Let H_i' , $i=1,2$

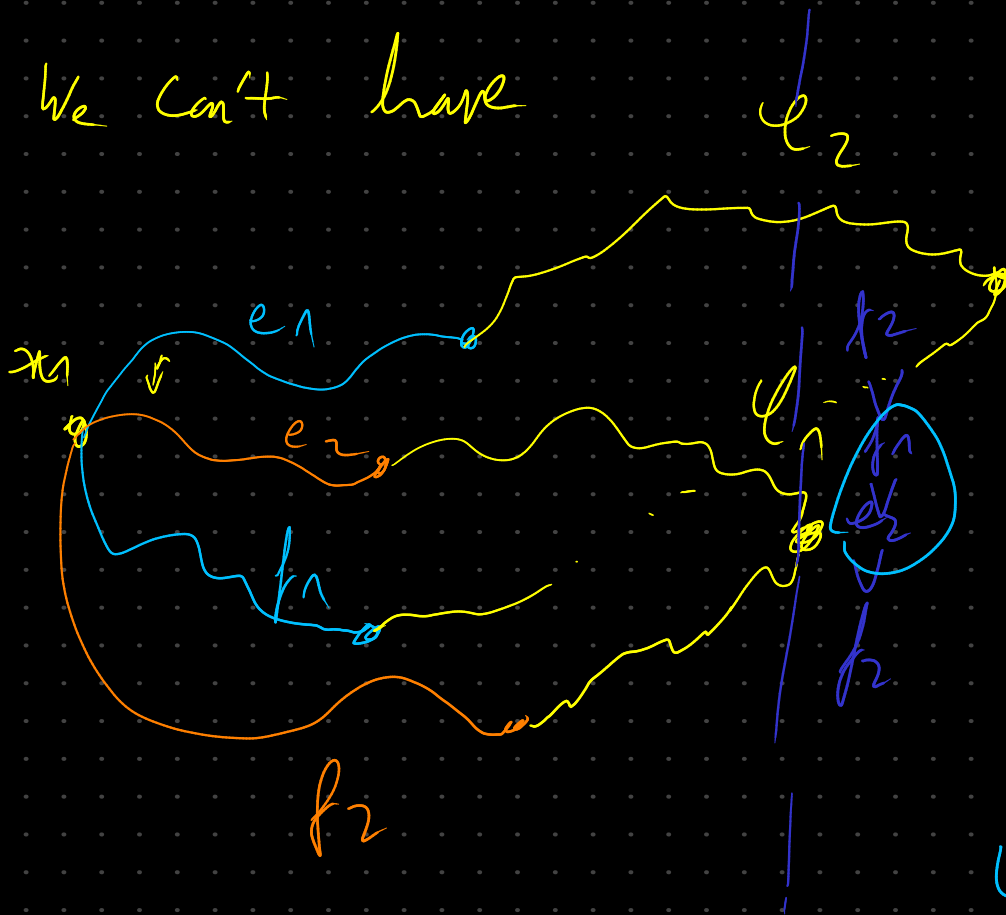
H_i , $i=1,2$



this is a comp by x_1

$H_2 = L[H_2 \cup x_1]$

We can't have



We have

$$f_1 <_{x_1} e_2$$

$\Downarrow$

$$l_{f_1} <_{\text{min}} l_{e_2}$$

but we have

$$l_{f_1} > l_{e_2}$$

$$\pi_{\text{in}}(n_{e_1}, n_{e_2})$$

there are

odd number of
crossings btw

l_{f_1} and l_{e_2}

for l_{f_1} and l_{e_1} they cross all other edges

evenly (no change of rel. pos. in $\pi_{\text{in}}(n_{e_1}, n_{e_2})$)

$\Rightarrow$ odd number of crossings btw l_1 and l_2

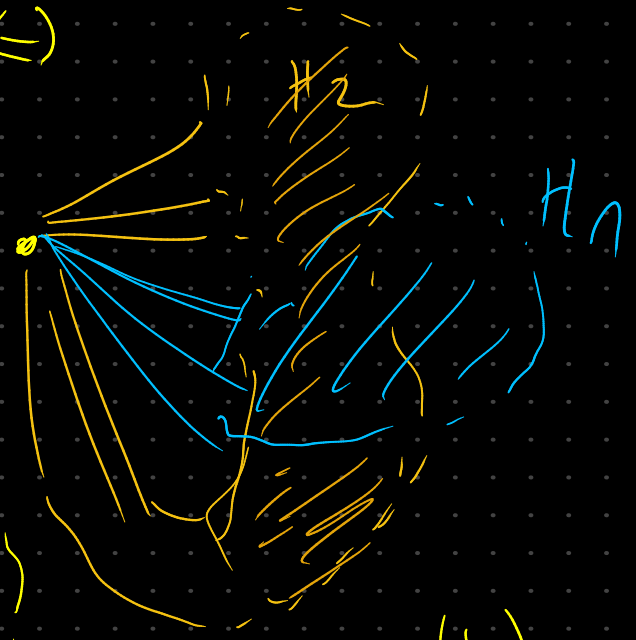
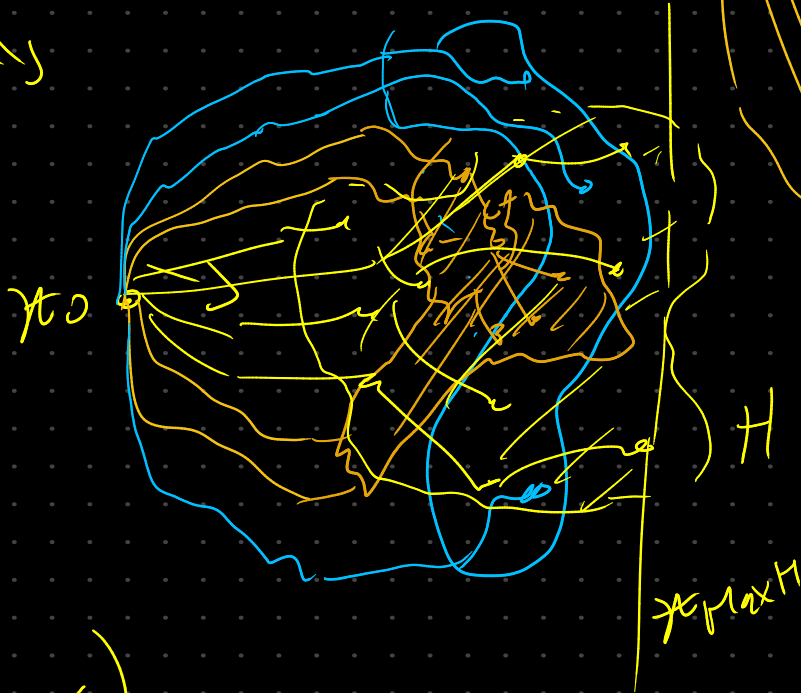
$\Rightarrow$ can't interleave $e_1 f_1$ with $e_2 f_2$.

that means we can put relative order b/w components

$$H_1 < H_2 \Leftrightarrow$$

let H be a min element in rel. to ' $<$ '

so $\Rightarrow$



$$x_0 \in V(H)$$

$$x_0 < x_{\max H}$$

$$(C, <) \quad S \subseteq C$$

max. el. of S

if $s \in S$

$$m < s \Rightarrow m \geq s$$

H is a maximal element.

$$x_{\max H} = \max_{v \in H} x(v)$$

in our case P is the set of the sub
w induced by $H_i = h[\{x_1\} \cup H'_i]$

and $S = P(S, \leq)$

H is a minimal element of S

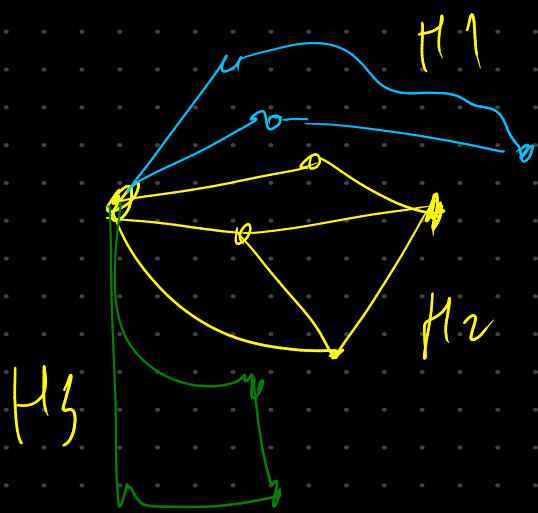
$\Leftrightarrow$ if $I \in S$ $I \leq H \Rightarrow H \leq I$

why having the right tot of h makes H also a maximal

aka (if $I \in S$ $I \geq H \Rightarrow H \geq I$)

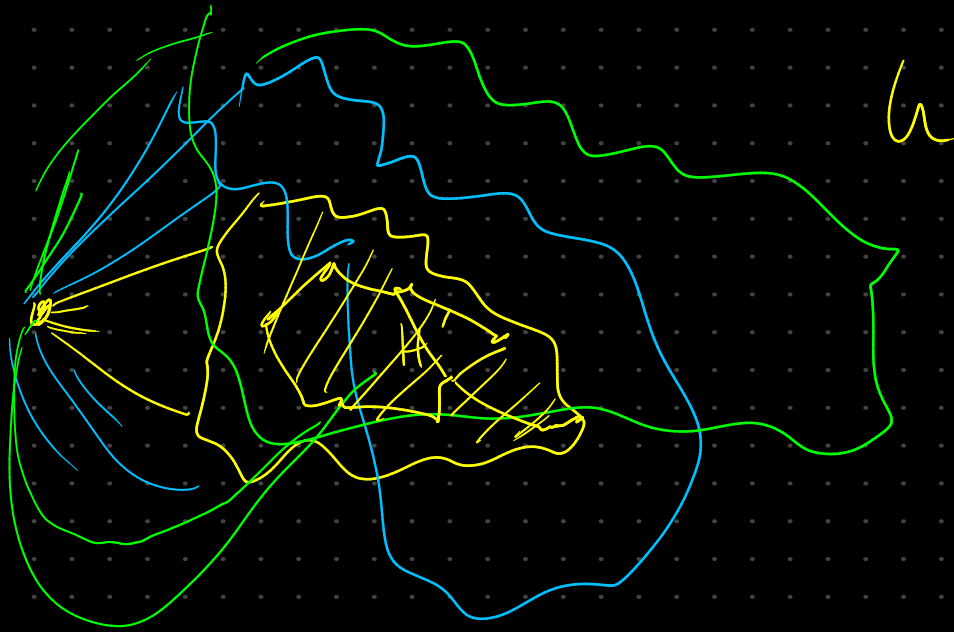
in case H is also max, every el of S equals to H ?

but why makes it maximal

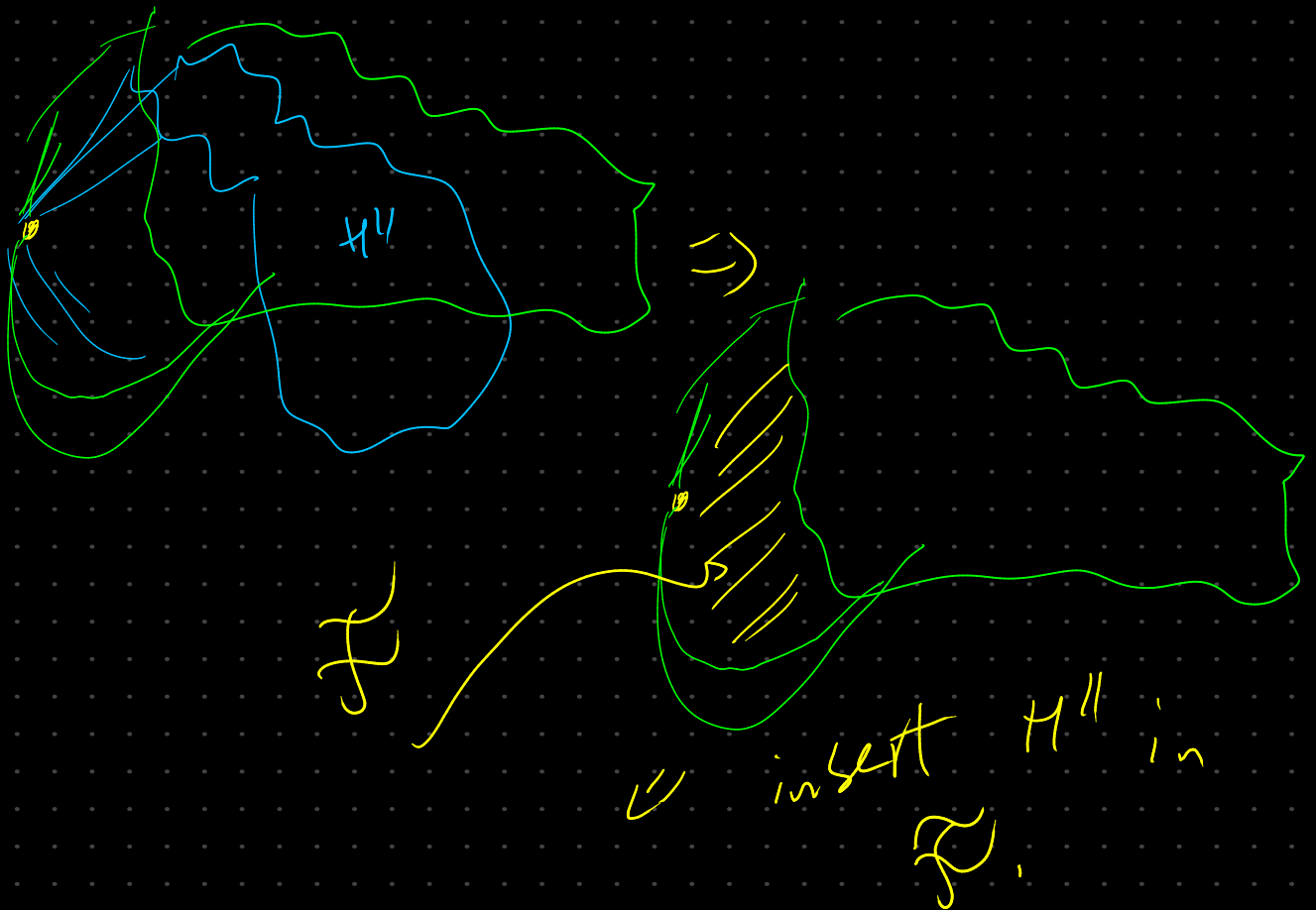


H_3 minimal
in $S = \{H_1, H_2, H_3\}$
in this set
all comp. comp
are min and max

Let $H' = H - \{x_n\}$, let's consider $G - V(H')$

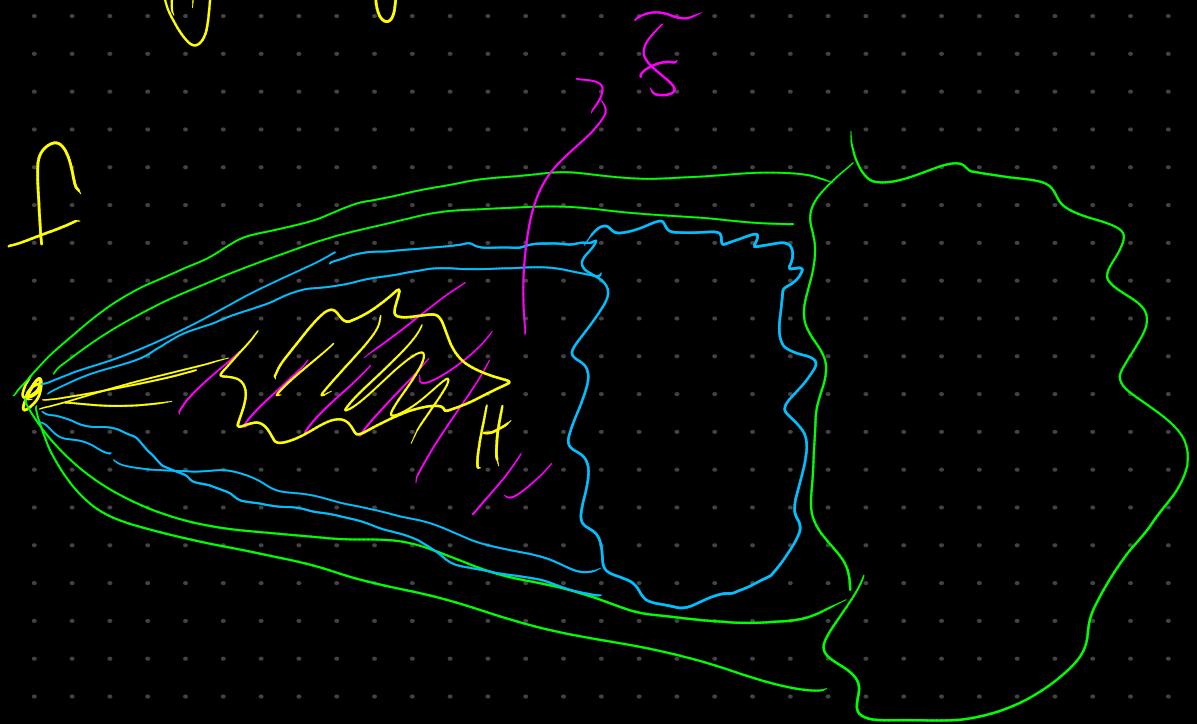


$\Downarrow$ $G - V(H')$





By induction



$\Rightarrow \Gamma$ is a x -monotone planar drawing
 having no edge in $\text{cap}(u)$
 gives an x -~~plan~~ embedding
 monotone



